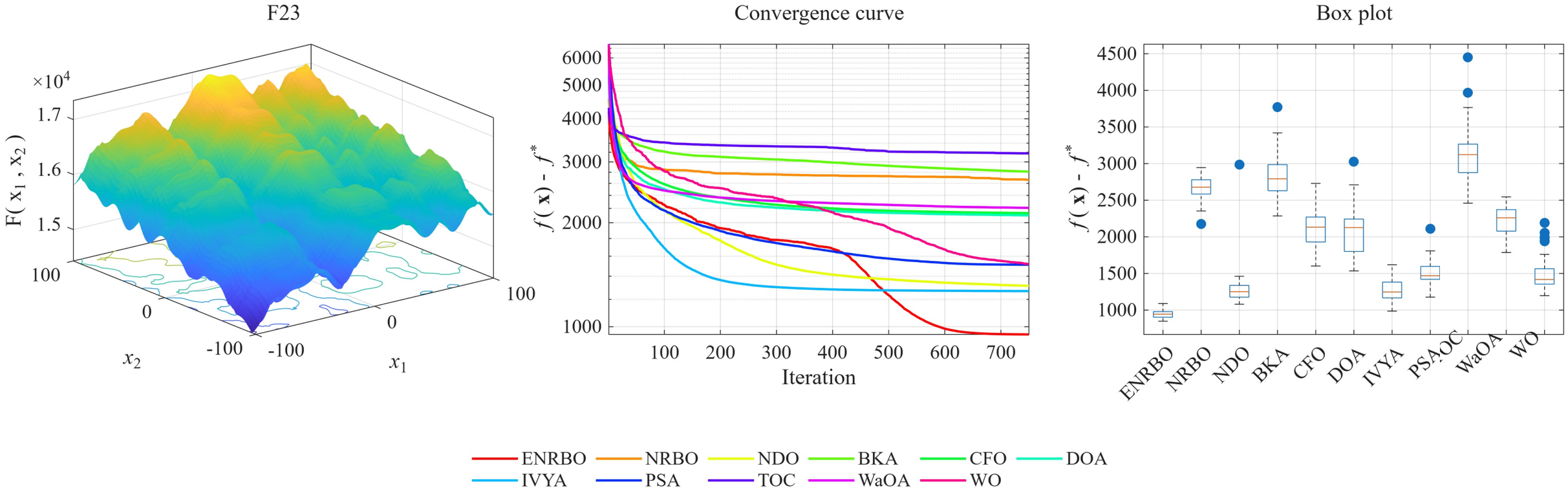

F23
×10^4
1.7
1.6
1.5
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
6000
5000
4000
3000
2000
1000
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
4500
4000
3500
3000
2500
2000
1500
1000
f(x) - f*
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
OC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO